

Dynamic Analysis of a Quarter-Car Suspension System

Objective: Study the normal contact force between the wheel and the road surface (tire-road adhesion).

Assumptions: Massless wheel; wheel follows the road profile; linear spring-damper suspension.

Variables: K = spring stiffness, c = damping coefficient, Δ = spring free length, X = vehicle vertical displacement, Y = wheel vertical displacement.

Spring force: $F_{\text{spring}} = -K(X - Y - \Delta)$

Damper force: $F_{\text{damper}} = -c(\dot{X} - \dot{Y})$

Equation of motion: $m\ddot{X} + c\dot{X} + KX = c\dot{Y} + KY + K\Delta - mg$

External excitation: $F_{\text{ext}}(t) = c\dot{Y} + KY + K\Delta - mg$

Road profile: $Y = A \sin(\omega t)$

Acceleration: $\ddot{X} = (-KX + F_{\text{ext}} - c\dot{X})/m$

Explicit Euler integration

$\ddot{X}(i+1) = (F_{\text{ext}}(i) - KX(i) - c\dot{X}(i))/m$

$\dot{X}(i+1) = \dot{X}(i) + \ddot{X}(i) \Delta t$

$X(i+1) = X(i) + \dot{X}(i+1) \Delta t$

The algorithm iteratively computes acceleration, updates velocity and displacement, and stores the results to simulate the vehicle response to sinusoidal road excitation.